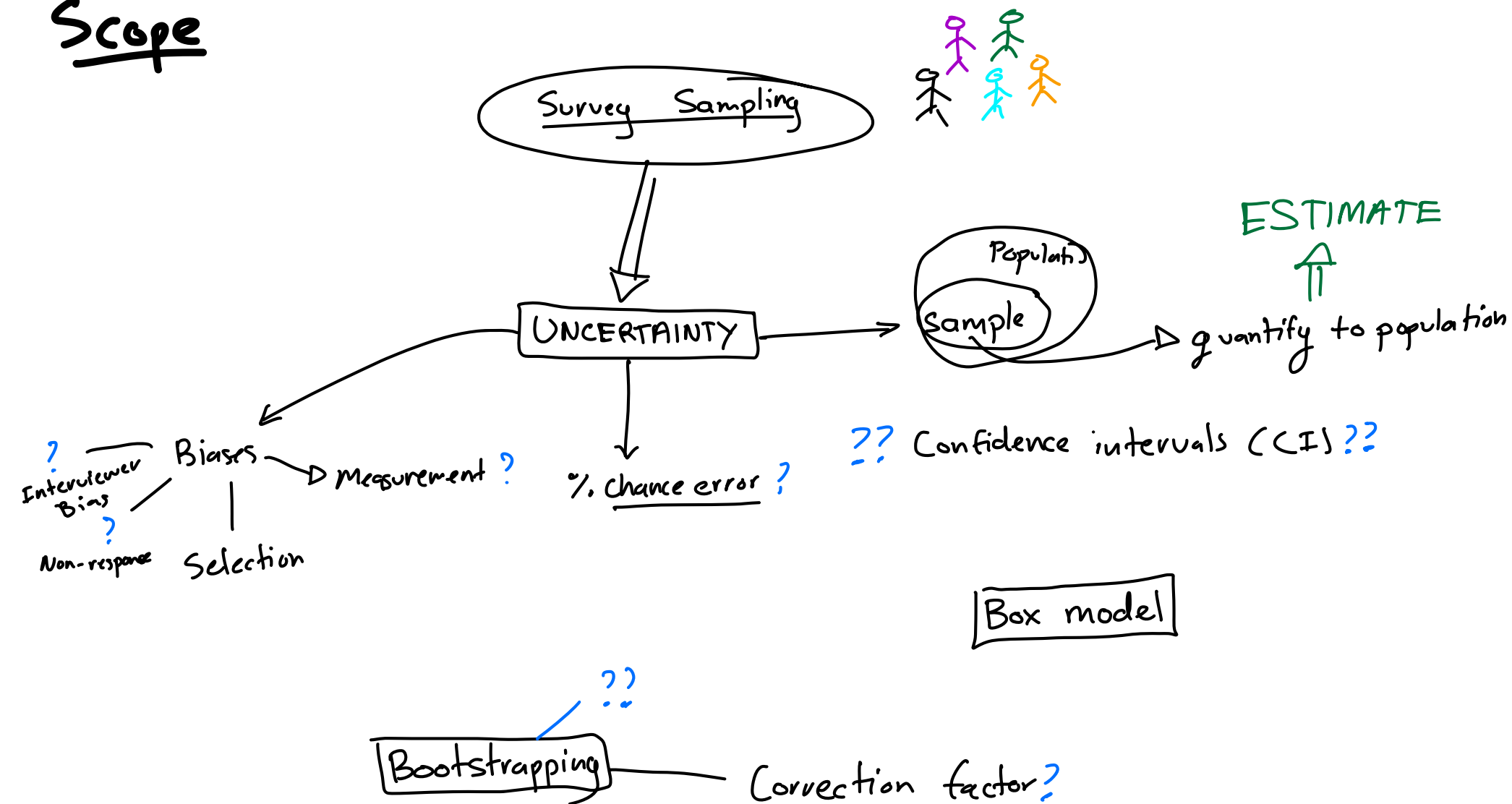
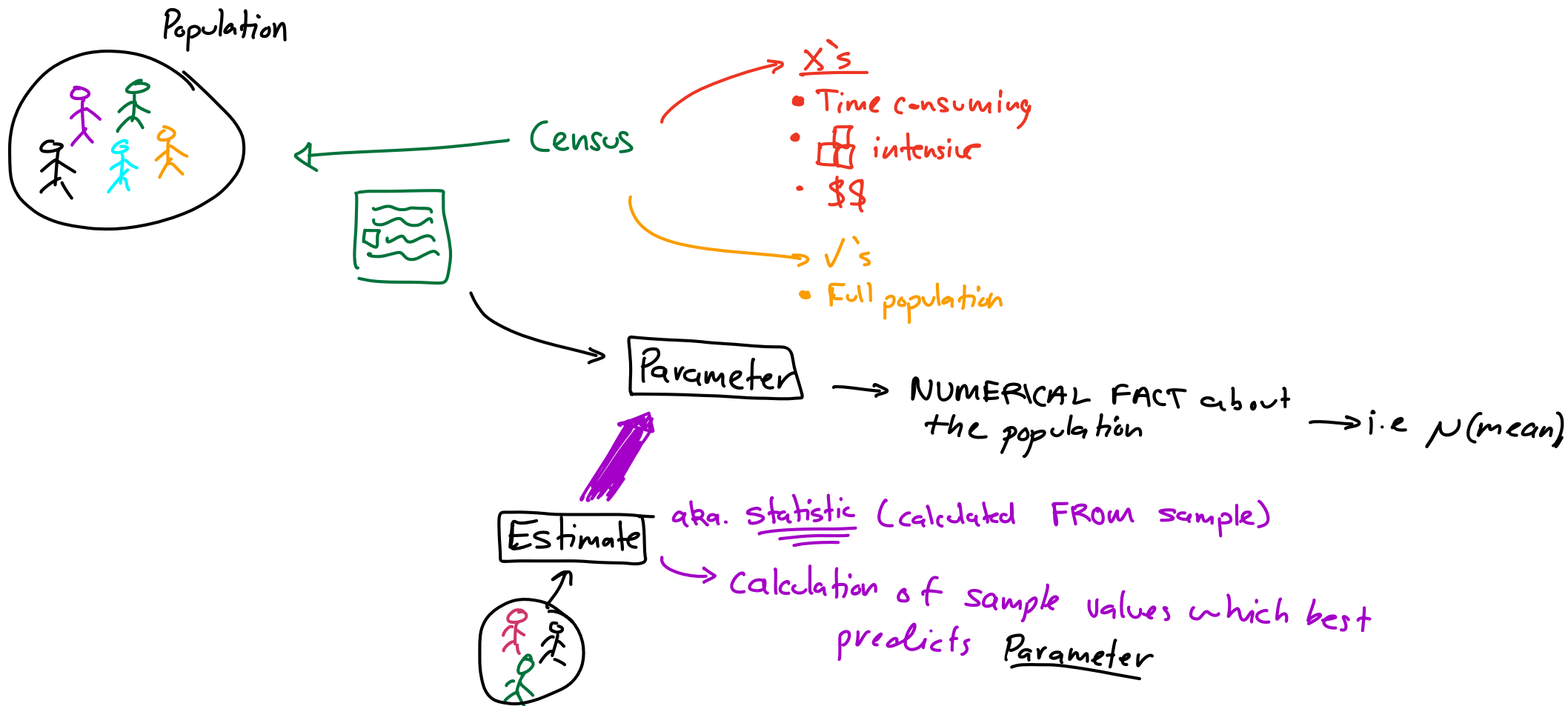


Scope





FACT

- Parameter

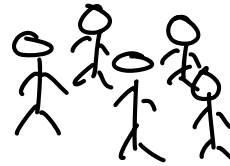
Sample // Not Accurate

- Estimate

(BIAS)



EXCLUSION



vs.

INCLUSION

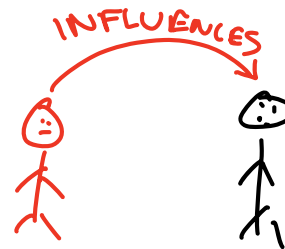


· Selection Bias →

· Non-response Bias →



· Interviewer Bias →



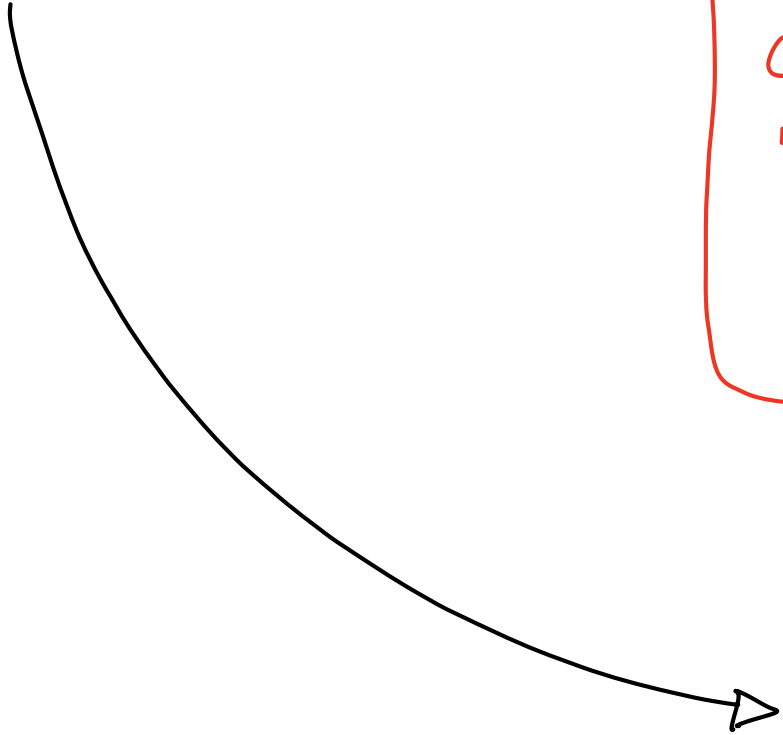
Form of measurement bias

Measurement

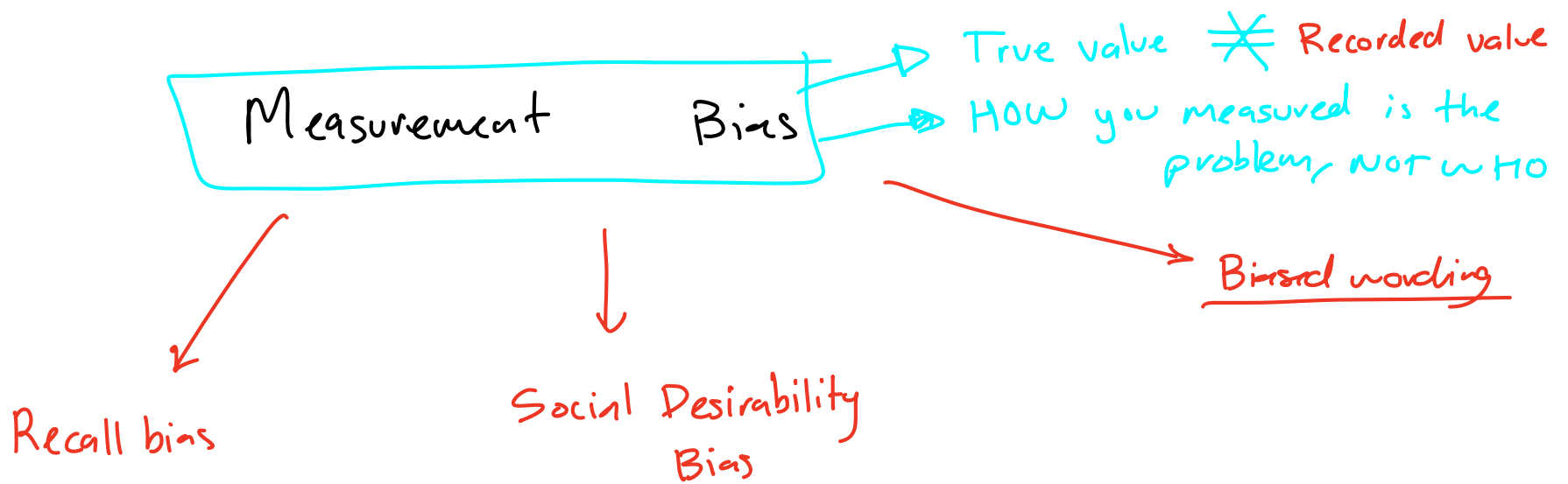
bias



(way question is
worded) → affects
answer



BIAS



GOOD SAMPLING

① ~~Interview~~

② Impartial selection

SAMPLING METHODS

① Simple Random Sampling

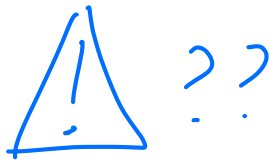
- drawing individuals from a population at RANDOM without replacement
- all % chance of being chosen is random

② Stratified Sampling



→ subpopulations } → sample is representative
(certain characteristics)

③ Multi-stage cluster sampling



- Sampling in stages

- Choose group

→ O sample in group

④ Quota Sampling

→ Non-probabilistic

→ assembled survey has a pre-determined no. of individuals for a SPECIFIC characteristic

→ unintentional bias by interviewees

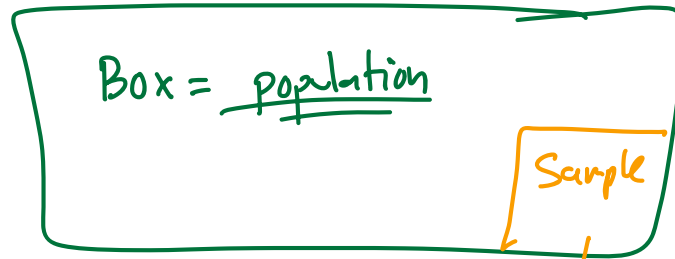
i.e. age
gender
ethnicity

⑤ Convenience Sampling

→ choosing subjects ∴ they are convenient

Modelling Proportions
of Samples
w/ BOX MODEL

The Box Model



"Tickets" → value in the population

Everytime you sample $\Rightarrow$ get a different result

Chance error

① Expected value = $\mu(\text{box})$

Chance error

② Standard error = $\frac{\text{Standard Deviation (Box)}}{\sqrt{n}}$
(how much sample varies)

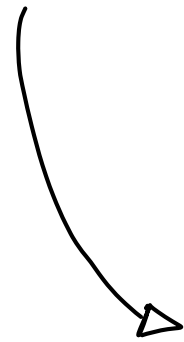
Proportion = $\mu(\text{sample})$

Central Limit Theorem (CLT)

$$\boxed{\text{Correction Factor (CF)}} = \sqrt{\frac{\text{population size} - \text{sample size}}{\text{population size} - 1}}$$

$$SE_{\text{w/out replacement}} = \text{correction factor} \times SE_{\text{w/ replacement}}$$

Confidence Intervals (CI) $\rightarrow$ range of plausible values for true population value



Sample proportion $\pm 1 \times SE$

CI $\rightarrow$ reliability of sampling method NOT % of parameter



- Fixed
- usually unknown
- NUMERICAL value of population

Goal of statistics

$\hookrightarrow$ estimate parameter

Bootstrapping



Bootstrapping is a method $\rightarrow$ repeatedly resample from your sample to estimate how a statistic behaves.

Bootstrapping Procedure

- ① Assume best representation of population
(with replacement)
- ② Re-sample your sample $\rightarrow$ make statistical prediction

$\downarrow$
represents DISTRIBUTION of μ 's

QUIZ

25%. Sydney people catch train